Vol 10 Chapter 4 — PDEs: Heat, Wave, Laplace

Keeper (author pass — deep math/physics revision)

2026-05-24 Sunday

Chapter 4 — PDEs: Heat, Wave, Laplace

Level 1 — one sentence

The three canonical second-order PDEs — heat (parabolic), wave (hyperbolic), Laplace (elliptic) — span the classification of linear second-order PDEs and underlie much of mathematical physics; the heat kernel on the BST substrate domain D_{IV}^5 is Paper #9's arithmetic-triangle subject.

Level 2 — graduate-physicist precision

4.1 The three canonical PDEs

Heat equation (parabolic): $\partial_t u = \alpha\nabla^2 u$. Models diffusion, heat conduction. Smoothing: rough initial data becomes smooth instantly.

Wave equation (hyperbolic): $\partial_t^2 u = c^2\nabla^2 u$. Finite propagation speed c. Preserves discontinuities along characteristics.

Laplace equation (elliptic): $\nabla^2 u = 0$. Harmonic functions: equal to mean value over surrounding spheres. Maximum principle: maxima on boundary.

4.2 Separation of variables

For PDE on rectangular domain with appropriate BCs: try $u(x, y, t) = X(x)Y(y)T(t)$, get separate ODEs (often Sturm-Liouville). Sum up eigenfunction expansion.

For heat equation on $[0, L]$ with $u(0) = u(L) = 0$: $u(x, t) = \sum_n c_n \sin(n\pi x/L)e^{-\alpha(n\pi/L)^2 t}$.

For wave equation: $u(x, t) = \sum_n [a_n \cos(\omega_n t) + b_n \sin(\omega_n t)] \sin(n\pi x/L)$ with $\omega_n = n\pi c/L$.

4.3 Heat kernel on D_IV^5

The substrate's heat kernel — solution of $\partial_\tau K = \Delta K$ on D_{IV}^5 — has the BST-canonical Seeley-DeWitt expansion (Vol 6 Ch 5, Paper #9):

$$K(z, z; \tau) \sim (4\pi\tau)^{-5} \sum_k a_k(z)\tau^k$$

with Paper #9's arithmetic triangle giving the coefficient structure: speaking-pair period $n_C = 5$, column rule (T531), two-source primes (T532), Kummer-analog conjecture (T533). Verified through $k = 20$.

4.4 Green's functions

For inhomogeneous PDEs $Lu = f$: Green's function G satisfies $LG(\vec{r}, \vec{r}') = \delta(\vec{r} - \vec{r}')$. Solution: $u(\vec{r}) = \int G(\vec{r}, \vec{r}') f(\vec{r}') d^n r'$.

For Poisson equation in 3D: $G(\vec{r}, \vec{r}') = -1/(4\pi|\vec{r} - \vec{r}'|) \to$ Coulomb-like Green's function.

4.5 Fundamental solutions

- Heat: $G_{\text{heat}}(\vec{r}, t) = (4\pi\alpha t)^{-n/2} e^{-r^2/(4\alpha t)}$ — Gaussian
- Wave (3D): $G_{\text{wave}}(\vec{r}, t) = \delta(r - ct)/(4\pi c^2 r)$ — sharp pulse on light cone
- Laplace (3D): $G = -1/(4\pi r)$

4.6 K-audit anchors

- Paper #9 "Arithmetic Triangle of Curved Space" (Vol 6 Ch 5)
- T531-T533: column rule, two-source primes, Kummer conjecture

Level 3 — 5th-grader accessibility

Three canonical PDEs: heat (diffusion), wave (light, sound), Laplace (electrostatics). Separation of variables: guess $u = X(x)Y(y)T(t)$, get separate ODEs, solve each by Sturm-Liouville. Heat kernel on D_{IV}^5 encodes substrate structure — Paper #9 found an arithmetic triangle pattern with BST primary $n_C = 5$ as the natural period, verified through 20 coefficient orders.

What comes next

Chapter 5 develops Fourier analysis.

Where to look this up

- Evans, *Partial Differential Equations*
- BST Paper #9; Vol 6 Ch 5